

MONSTER PACKS

Game Design Document

com.sergivm.monsterpacks

Version 0.2 — Expanded & Refined

Single source of truth. Includes all design, values, and technical decisions.

1. Technical Stack

The project targets Android exclusively, written in Kotlin with Jetpack Compose. No Java, no LiveData, no GlobalScope. All async operations use Coroutines + Flow/StateFlow.

Technology	Version	Notes
Kotlin	1.9.22	Primary language. No Java.
AGP	8.3.0	Do not update without reviewing KSP compatibility.
KSP	1.9.22-1.0.17	Must match Kotlin version exactly.
minSdk	28 (Android 9)	Minimum supported.
targetSdk / compileSdk	34	Current target.
Jetpack Compose BOM	2024.02.00	Manages Compose versions internally.
Hilt	2.50	Dependency injection.
Room	2.6.1	Local persistence.
Coroutines	1.7.3	Game loop and async operations.
Gson	2.10.1	GameState serialisation to JSON for Room.
DataStore	1.0.0	Lightweight config and preferences.
Navigation Compose	2.7.7	Screen navigation.

1.1 Architecture — Clean Architecture (Strict)

- domain/model/ — All entities and models. Each in its own file.
- domain/engine/ — GameEngine: pure logic, zero Android or Compose imports.
- domain/usecase/ — Optional use cases for complex logic.
- data/db/ — Room entities, DAOs, Database.
- data/repository/ — GameRepository: load and save game state.
- presentation/viewmodel/ — Hilt ViewModels connecting engine to UI.
- presentation/screen/ — Compose screens.
- presentation/ui/theme/ — Theme, colours, typography.
- app/di/ — Hilt modules.

⚠ GameEngine must NOT import anything from Android or Compose. Pure logic, fully testable with JUnit.

⚠ No business logic in ViewModels. No logic in Compose. Never use LiveData — Flow and StateFlow throughout. Never block main thread with Room — always Dispatchers.IO.

2. Game Overview

Monster Packs is an offline mobile card-collecting game. Players open packs to discover creatures and complete collections. No internet connection is required. The game features colorful, epic visual design with elegant and detailed aesthetics — rich animations, vibrant colours, and a non-minimalist art style.

2.1 Core Loop

- Open the free Basic Pack — unlimited, no cooldown.
- Collect cards, track new discoveries and duplicates.
- Earn Coins and Gems from opened cards.
- Spend resources on Special Packs or upgrades in the Shop.
- Complete collections to unlock cosmetic rewards (future).

2.2 Navigation — Bottom Tab Bar

Four main tabs, in order:

Tab	Icon (placeholder)	Description
Pack		Main free pack opening screen.
Collection		View all cards, unlocked and locked.
Shop		Buy special packs, upgrades.
Settings		Language, Google Play save, legal links.

Additional tabs may be added in future versions.

3. Models

Cards are pre-defined static entities. Opening a pack does not create new Card entities — it reads from the existing pool and updates the player's copy count. Pack opening sessions are not persisted; only the resulting card counts matter.

3.1 Card

Field	Type	Description
id	Int	Unique identifier.
name	String	Display name of the card.
type	CardType (enum)	Determines background colour and thematic family.
rarity	Rarity (enum)	Determines card shape/silhouette and drop probability.
collection	Collection (ref)	The collection this card belongs to.
imageRes	String / ResId	Placeholder image or final illustration reference.
coinReward	Int	Coins awarded when this card is obtained.
gemReward	Int	Gems awarded (0 for Common/Rare/Epic).
copies	Int	Player-owned copies. 0 = locked (shown as silhouette).

3.2 Pack (PackDefinition)

Packs are defined statically. A PackDefinition describes all the rules for a pack type — it is not a runtime entity.

Field	Type	Description
id	String	Unique pack identifier (e.g. "pack_basic", "pack_fire").
name	String	Display name.
type	PackType (enum)	Category: BASIC, TYPE_THEMED, RARITY_BOOSTED, LUCKY, REWARD.
collection	Collection (ref)	Card pool source.
cardPool	List<CardType>?	If TYPE_THEMED: restricts pool to these card types.
slotRules	List<SlotRule>	Per-slot guaranteed rarity or roll distribution.

upgradeLevel	Int	Current upgrade tier (for Basic and Reward packs).
cost	PackCost?	Null for free packs; otherwise Coins or Gems amount.
designRes	String / ResId	Visual design asset — changes per pack type.

3.3 Collection

Field	Type	Description
id	String	Unique identifier.
name	String	Display name.
cards	List<Card>	All cards belonging to this collection.

3.4 PlayerState

Field	Type	Description
coins	Long	Current coin balance.
gems	Long	Current gem balance.
cardCopies	Map<Int, Int>	cardId → copy count.
rewardPackReadyAt	Long?	Epoch ms when next Bonus Pack is ready.
basicPackUpgradeLevel	Int	Upgrade tier for the Basic Pack.
rewardPackUpgradeLevel	Int	Upgrade tier for the Bonus Pack.

4. Card Types

Each type defines a thematic family and a unique background colour for its cards. There are 10 types in the initial collection. Each type will have between 10 and 20 cards.

Type	Theme / Family	Colour (hex)	Example Creatures
Infernal	Fire, volcanoes, magma, hell	#C0392B (deep red)	Fire Dragon, Magma Golem, Hellhound, Lava Serpent, Ember Phoenix
Abyssal	Deep sea, darkness, pressure	#1A3A5C (dark navy)	Kraken, Leviathan, Abyssal Eel, Bioluminescent Jellyfish, Deep Shark
Spectral	Death, dark magic, undead	#4A235A (dark purple)	Banshee, Wraith, Shadow Vampire, Lich, Plague Bat
Celestial	Stars, cosmos, divine light	#D4AC0D (gold)	Solar Griffin, Stardust Unicorn, Meteor Golem, Nebula Serpent, Comet Fox
Storm	Lightning, wind, tempests	#2874A6 (electric blue)	Thunderbird, Cyclone Drake, Storm Giant, Lightning Eel, Gale Hawk
Verdant	Forest, nature, plants, earth	#1E8449 (forest green)	Treant, Vine Hydra, Mushroom Golem, Thorn Wolf, Mossy Tortoise
Frost	Ice, tundra, arctic creatures	#AED6F1 (ice blue)	Frost Giant, Glacier Wyrms, Permafrost Yeti, Arctic Kitsune, Ice Sphinx
Arcane	Pure magic, runes, constructs	#7D3C98 (violet)	Runic Golem, Mana Wisp, Arcane Chimera, Spellbound Gargoyle, Ether Serpent
Savage	Wild beasts, primal instinct	#D35400 (burnt orange)	Dire Bear, Saber-Tooth Tiger, Giant Boar, Razorback Wolf, Primal Raptor
Mythic	Gods, legends, ancient myths	#B7950B (antique gold)	Minotaur, Medusa, Cerberus, Cyclops, Sphinx

i These 10 types replace the initial 4 from v0.1. The original concepts (Spectral, Volador, Fuego, Agua) have been absorbed and expanded into Spectral, Storm, Infernal, and Abyssal respectively, preserving the spirit of the original ideas.

5. Card Examples per Type (Initial Proposals)

Each type will have 10–20 cards. Below are 8 initial card name proposals per type to serve as a design starting point. Names are in English; Spanish localisation to be added.

5.1 Infernal

- Ignis the Flame Drake
- Pyro Hellhound
- Lava Golem Prime
- Ember Phoenix
- Molten Serpent
- Cinderclaw Wyvern
- Magma Titan
- Scorchfang

5.2 Abyssal

- The Kraken
- Leviathan of the Deep
- Voidfin Shark
- Lurker Eel
- Brine Jellyfish
- Deep Horror
- Abyssal Manta
- Pressure Crab

5.3 Spectral

- The Wraith
- Shadow Bat
- Pale Banshee
- Lich of the Abyss
- Specter Wolf
- Duskborn Vampire
- Grave Gargoyle
- Phantom Cat

5.4 Celestial

- Solar Griffin
- Stardust Unicorn

- Nebula Serpent
- Comet Fox
- Meteor Golem
- Astral Pegasus
- Lunar Moth
- Dawn Phoenix

5.5 Storm

- Thunderbird
- Cyclone Drake
- Gale Hawk
- Lightning Eel
- Storm Giant
- Tempest Lynx
- Hurricane Serpent
- Squall Bat

5.6 Verdant

- Elder Treant
- Vine Hydra
- Thorn Wolf
- Mossy Tortoise
- Mushroom Golem
- Bramble Bear
- Petal Sprite
- Root Serpent

5.7 Frost

- Glacier Wurm
- Permafrost Yeti
- Arctic Kitsune
- Ice Sphinx
- Frost Giant
- Snowdrift Owl
- Blizzard Serpent
- Tundra Wolf

5.8 Arcane

- Runic Golem
- Mana Wisp
- Arcane Chimera
- Ether Serpent
- Spellbound Gargoyle
- Glyph Drake
- Hex Cat
- Construct Prime

5.9 Savage

- Dire Bear
- Saber-Tooth Tiger
- Giant Boar
- Razorback Wolf
- Primal Raptor
- Ironhide Rhino
- Feral Lynx
- Bone Crusher Ape

5.10 Mythic

- The Minotaur
- Medusa
- Cerberus
- The Cyclops
- Sphinx of Riddles
- Hydra
- Chimera
- Manticore

6. Rarities

Rarity defines drop probability, card silhouette/shape, coin reward, gem reward, and opening animation intensity. Higher rarities have more spectacular reveal animations.

Rarity	Coin Reward	Gem Reward	Base Drop %	Animation Tier
Common	1	0	~60% of free rolls	Simple fade-in
Rare	2	0	~35% of free rolls	Glow pulse
Epic	10	0	5%	Particle burst
Special	100	10	1%	Full-screen shatter
Legendary	500	50	0.1%	Epic cinematic
God	2000	200	0.01%	Full legendary sequence

i 'Base Drop %' applies to free roll slots (Slots 4 and 5 of the Basic Pack). Slots 1–2 are always Common; Slot 3 is always Rare.

7. Packs

7.1 Basic Pack (Free, Unlimited)

The core of the game. No cooldown, no limit. Each pack reveals 5 cards one by one (tap to reveal each). After all 5 are shown, a summary screen appears with a Save button to persist results.

Slot	Rule
Slot 1	Always Common
Slot 2	Always Common
Slot 3	Always Rare (exactly Rare, not higher)
Slot 4	Free roll: ~58% Common, ~36% Rare, 5% Epic, 1% Special, 0.1% Legendary, 0.01% God
Slot 5	Same free roll as Slot 4

i After determining rarity, the game rolls within the full card pool of that rarity to select which specific card appears.

7.2 Bonus Pack (Free, Cooldown-Based)

Available in the Shop tab. Generates automatically every 10 minutes. Gives 3 cards + 5 Gems. Multiple packs can accumulate up to a storage cap. Both the cooldown and the storage cap can be upgraded.

Field	Default Value
Cards per pack	3
Gems per pack	5
Cooldown	10 minutes
Max stored packs	1 (upgradeable)

7.3 Special Packs (Paid)

Purchased with Coins or Gems in the Shop. Every Special Pack has a unique visual design. Pack designs change visually depending on the pack type.

7.3.1 Type-Themed Packs

One pack per Card Type (10 total). Restricts the card pool to that type only. Standard rarity distribution applies.

7.3.2 Rarity-Boosted Packs

One pack per boosted rarity tier. All 5 slots use an elevated probability table — e.g. an Epic Pack guarantees at minimum Epic rarity on all slots.

7.3.3 Lucky Packs

All cards are at minimum a defined minimum rarity. Variants:

- Lucky Epic Pack — all cards Epic or above.
- Lucky Special Pack — all cards Special or above.
- Lucky Legendary Pack — all cards Legendary or above.

7.4 Pack Upgrade Event (Surprise Pack)

During Basic Pack opening, there is a low-probability chance that the pack 'evolves' into a Special Pack before the first card is revealed. The trigger point is when the player taps to open — instead of the standard open animation, a special 'evolution' animation plays, upgrading the pack visually and improving the odds for that session.

Event detail	Value / Note
Trigger moment	On player tap to open — before first card reveal
Animation	'Evolution' animation upgrades pack appearance before opening
Base probability	To be tuned; suggested starting value: 0.5%
Resulting pack type	Random Special Pack or Rarity-Boosted Pack

8. Resources

8.1 Coins

The common resource. Earned by opening cards. Used to buy certain Special Packs and to upgrade the Basic Pack and Bonus Pack in the Shop.

Source	Amount
Common card obtained	+1 coin
Rare card obtained	+2 coins
Epic card obtained	+10 coins
Special card obtained	+100 coins
Legendary card obtained	+500 coins
God card obtained	+2000 coins

Note: Coins are always awarded regardless of whether the card is new or a duplicate.

8.2 Gems

The premium resource, earned slowly. Used to buy premium Special Packs and upgrades.

Source	Amount
Special card obtained	+10 gems
Legendary card obtained	+50 gems
God card obtained	+200 gems
Bonus Pack opened	+5 gems (base)

8.3 Duplicates

When a card the player already owns appears in a pack, it is still awarded its full Coin and Gem reward. The copy count increments normally. A future version will introduce a 'fuse' mechanic to convert excess duplicates into resources.

i Remember for future update: implement a Fuse system to trade duplicates for Coins, Gems, or other rewards.

9. Card Opening UX Flow

The visual and interaction design for pack opening is a key pillar of the game's feel.

Step	Description
1. Pack screen	Player sees the pack. Pack design reflects its type. Button: 'Open'.
2. (Optional) Evolution event	On tap, low-probability check. If triggered: evolution animation upgrades pack before proceeding.
3. Pack open animation	Pack-specific opening animation plays. 5 card slots appear face-down.
4. Card reveal (x5)	Player taps each card to flip it. Higher rarity = more spectacular flip animation and particle effects.
5. 'New!' / Copies badge	If the card is new: 'New!' tag appears. If duplicate: shows current copy count (e.g. x3).
6. Summary screen	All 5 cards displayed together. Save button persists results to PlayerState.

10. Collection Screen

The Collection tab shows all cards for a given collection, both unlocked and locked.

- Unlocked cards: show full illustration, name, type, rarity, copy count.
- Locked cards: shown as a black silhouette of the card shape. Name may be hidden or shown (TBD).
- Filter and sort options (by type, rarity, owned/unowned) — proposed for v1.
- Progress indicator: X / Y cards collected in this collection.

11. Settings

Setting	Options / Notes
Language	English / Español. More languages in future.
Google Play Save	Connect account to save progress to the cloud.
Privacy Policy	Link to external URL.
Terms of Service / Legal	Link to external URL or in-app screen.

12. Visual & UX Style

The game has a colorful, epic visual identity. The style is modern and elegant with rich detail — the opposite of minimalist.

- Rich colour palettes per card type — each type has a distinct, vibrant background.
- Card shapes (silhouettes) differ per rarity — higher rarity = more elaborate frame/shape.
- Opening animations scale with rarity — from simple fades to full-screen cinematic sequences.
- Pack designs are unique per type — the pack art changes to match its theme.
- Dark background recommended as base — lets card colours and effects pop.
- Fonts: bold, modern, with fantasy undertones. Not serif-heavy; clean but impactful.

13. Future Features (Roadmap)

The following features are planned for future versions and should be considered during architecture design to avoid rework.

Feature	Notes
Pack Upgrades	Upgradeable Basic Pack and Bonus Pack — improve cards, cooldowns, gem yield.
Fuse / Duplicates	Trade duplicate cards for Coins, Gems, or crafting materials.
Ad Rewards	Optional video ads to earn packs, gems, or coins.
Picks	Choose 1 card from a selection of 5 randomly presented options.
Scratch Cards	Scratch-and-win mechanic for bonus rewards.
More Collections	New card pools and themes beyond the initial collection.
Daily & Weekly Missions	Short-term goals for consistent engagement.
Infinite Achievements	Long-term achievement system with scaling rewards.
More Tabs	Additional navigation tabs as features are added.

i First public release will include all of the above features. Early internal builds may omit them.

Changelog

Version	Date	Summary
0.1	Initial	First idea draft — base concept, stack, initial types and rarities.
0.2	2026-04-12	Full expansion: 10 card types proposed, card examples, detailed pack system, SlotRule model, Pack Upgrade Event, Bonus Pack naming, resource table, duplicate policy, future roadmap, visual style guide.

FIXES AND IMPROVEMENTS:

- All in Bonus Pack is upgradeable (cards, gems, time and stored count)
- A pack can't contain more than one same card
- The pack has to be ordered in rarity, so the 4th card can't never be a higher rarity than the 5th. Is it better to "reorder" the cards before showing them?
- Your card ideas are honestly incredible. I will add that I don't want to limit the cards to "fantastical" creatures, we can include plants, objects, real animals, landscapes, etc.
- We need to define the cards rarity of those created. From common to epic. The special and legendary will be "redesigns" of the main cards. For example, the Molten Serpent will have the "rare" or "epic" variant, and the "special" variant with a better image and card overall looking. Another example; Cinderclaw Wyvern will have the epic variant and the legendary variant, with a special layout and a full-art style. The common cards will not have special or legendary versions. Some rare cards can have special versions. Some epic cards can have legendary versions. Then, once for each type, a God card will be defined from the common, rare or epic cards. This god card should be a representative card of that type.
- Are Rare Boosted Packs and Lucky Packs the same? Also, we have to take care and define the probability of an evolving pack to become of each type. The Legendary Packs can't have the same probability as the Epic pack.
- Not all types need to have the same amount of cards, This way we can create the illusion of some types being more "exclusives". But each type needs at least 12 cards, and max 20.
- I am thinking of also having a "User" level. Each pack will grant a defined XP, that will not vary with the obtained cards, and this level will unlock the possibility of buying the upgrades. For example, the level 1 user needs 100xp points to upgrade to level 2. Each pack gives 10xp (this will be upgradeable too!). So when he opens 10 packs, he is now level 2 and can upgrade, for example, the bonus pack cooldown by 30 seconds. This is all an example, we will have to define it.